



Miniprep (Plasmid Extraction)

Designing Life with AI

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Description

Small-scale procedure to isolate, extract and purify plasmid DNA from a bacterium. Usually made as a stock for further experiments like transformation of a new strain. From the Promega Mini Prep Kit:

<https://ch.promega.com/products/nucleicacidextraction/plasmidpurification/pureyield-plasmid-miniprep-system>

Steps

I – Lysate preparation

1. Centrifuge 1.5 ml of bacterial culture for 1 minute at 13000 rpm in a microcentrifuge.
2. Discard the supernatant.
3. Add 600 µl of MilliQ water to the cell pellet, and resuspend completely.
4. Add 100 µl of Cell Lysis Buffer (Blue), and mix by inverting the tube 6 times.
5. Incubate 2 minutes at room temperature.
6. Add 350 µl of cold Neutralization Solution (stored in fridge), and mix thoroughly by inverting.
7. Centrifuge at maximum speed for 3 minutes.

II – Transfer in column

1. Place the minicolumn into a Collection Tube and transfer the supernatant (~900 µl) without disturbing the cell pellet.
2. Centrifuge 1 minute at 13000 rpm.
3. Discard the flowthrough, and place the minicolumn into the same collection tube.

III – Wash

1. Add 200 µl of Endotoxin Removal Wash (ERB) to the minicolumn. Centrifuge at 13000 rpm for 30 seconds.
2. Add 400 µl of Column Wash Solution (CWC) to the minicolumn. Centrifuge at 13000 rpm for 30 seconds.

IV – Elution

1. Transfer the minicolumn to a clean 1.5 ml microcentrifuge tube (label it), then add 30 µl of Elution Buffer directly to the minicolumn matrix.
2. Let stand for 1 minute at room temperature.
3. Centrifuge for 1 minute to elute the plasmid DNA.

V – Verification

1. Measure DNA concentration and DNA / RNA purity with a nanodrop machine (use water as the blank).

If the DNA solution still has a poor purity, perform another washing step.

VI – Storage

1. Store the tube at 4 or -20 °C depending on usage.